

ICUMS Simulator Administrator Manual

Complete course creation field reference and publishing workflow

Current system setup September 2026

Training simulator only - not official ICUMS

Purpose. This manual tells administrators and instructors exactly where to begin, what to enter in each content field, how conditional fields behave, and how to publish a complete course from theory through the final assessment and practical competency stage.

The recommended starting point is Instructor workspace > Course Builder. Django Admin is the advanced workspace for detailed configuration, document production, enrolment, evidence review, and corrections that the guided builder does not yet expose.

Contents

- 1 Roles and workspaces
- 2 Complete course setup sequence
- 3 Guided Course Builder fields
- 4 Theory lessons and questions
- 5 Theory assessments
- 6 Practical simulator flows
- 7 Fictitious shipping documents
- 8 Practical rubrics and competency assessment
- 9 Publishing and enrolment
- 10 Learner flow and completion
- 11 Advanced Django Admin reference
- 12 Final quality checklist and troubleshooting

Roles and workspaces

Use the guided Course Builder for normal course authoring. Use Django Admin when a relationship, version, rubric rule, document, enrolment, or evidence record needs advanced control.

Workspace	Primary users	Use it for
Instructor dashboard and Course Builder	Instructor, Administrator	Create draft courses, modules, theory lessons, lesson questions, module assessments, guided practical flows, review, and publish.
Django Admin	Administrator and trusted advanced instructor	Users, roles, enrolments, detailed question versions, final theory assessment, scenario state machines, rubrics, fictitious BL/invoice/packing documents, completion policy, and read-only evidence.
Student dashboard	Student	Theory roadmap, lesson reading, knowledge checks, assessments, orientation, practical scenarios, results, and certificates.
Simulator workspace	Student	Run the programmed practical flow. It saves attempts and actions without changing the course definition.
Practical portal	Student	Access Cargo, Clearance, Single Window, and scenario training areas. Return to Workspace goes back to the learning workspace.

Complete course setup sequence

Follow this order. It prevents broken links, missing questions, inaccessible assessments, and courses that cannot be published.

1. Prepare the course outline: course name, module order, lesson outcomes, theory content, questions, practical steps, pass marks, and required learner documents.
2. Open Instructor workspace > Course Builder. Choose Theory when the course contains reading and questions only, or Theory and Practical when it also includes simulated flows.
3. Create the course. The system creates Programme version 1 in Draft status.
4. Add modules in teaching order. Add each lesson, its reading blocks, knowledge-check question, answers, and explanation.
5. Allow the lesson builder to add each lesson question to its module assessment, or configure assessment items manually in Django Admin.
6. For a combined course, add one or more guided practicals to the relevant module. Enter the learner-facing steps in their correct sequence.
7. Create the required fictitious Bill of Lading, Commercial Invoice, Proforma Invoice, and Packing List in Django Admin and connect each to the correct scenario version.
8. Create or verify the final theory assessment in Django Admin. It belongs to the programme version, not to one module.
9. For assessed practical work, confirm the scenario purpose, assistance mode, rubric, criteria, pass percentage, and mandatory rules.
10. Open Review course. Resolve every issue. Publish only after checking content, answers, practical order, documents, and assessment settings.
11. Create the student account and enrol the student into the published programme version. The Student role is added automatically when enrolment is created through the user page.
12. Test with a student account: disclaimer, theory lessons, module assessment, final theory, orientation, practical, evaluation, and certificate rules.

Guided Course Builder fields

Create course

Field	What to enter	Behaviour and rules
Course name	A clear learner-facing name, such as International Trade and Customs	Creates a new Programme and Draft Programme Version 1. A unique internal code is generated automatically.

Add module

Field	What to enter	Behaviour and rules
Module title	One major topic or learning stage	Used for navigation, assessment naming, and learner progress.
Description	What learners should understand or be able to do	Optional but recommended. It appears as module context.
Order	Created automatically from the module sequence	Advanced Admin can adjust it. Lower numbers appear first.
Prerequisite	Earlier module that must be completed first	Advanced Admin field. Leave blank when modules are independent.

Add theory lesson

Field	What to enter	Behaviour and rules
Lesson title	Specific subject of the lesson	Generates a unique lesson slug within the module.
Summary	Short overview of what the lesson teaches	Displayed before or beside lesson content.
Key concept heading	Heading for the main reading section	Creates the first Text content block.
Teaching content	Complete theory explanation in learner-friendly language	Required. This is reading material, not a simulator step.
Example heading and body	A worked example or realistic fictional illustration	Optional. Both belong to an Example content block.
Important note heading and body	Warning, rule, exception, or reminder	Optional. Creates a Notice content block.
Knowledge-check question	One clear question testing the lesson	The guided builder creates a Single Choice Question Version 1.
Correct answer	The one correct option	Saved as the first option and marked correct.
Incorrect answers	Plausible but clearly incorrect alternatives	Two are required and a third is optional. Do not use trick answers.
Explanation	Why the correct answer is correct	Shown after the learner answers. Explain the rule, not merely the letter or option.
Also add to module assessment	Normally leave selected	Creates or reuses the module assessment and adds this question to it.

Lesson publication behaviour. Lessons, content blocks, question versions, module assessments, and practicals remain drafts while the course is being built. Publishing the course publishes the connected modules, lessons, lesson question versions, assessments, practical versions, and practical rubric versions.

Theory lessons and questions

Theory is reading with embedded knowledge checks and formal assessments. It does not contain simulator actions. Each lesson should have at least one teaching block and at least one knowledge check before publication.

Advanced lesson fields

Field	What to enter	Behaviour and rules
Module	The draft module containing the lesson	Must belong to the draft programme version.
Slug	Short lowercase identifier	Unique inside the module. The guided builder creates it automatically.
Order	1, 2, 3 and so on	Controls lesson sequence.
Is published	Selected only when ready	The guided course publish action sets it. Published programme content is protected from casual edits.
Content block kind	Text, Example, or Notice	Changes how the learner-facing section is presented.
After block	The content block after which a check appears	Must belong to the same lesson.

Question and answer fields

Field	What to enter	Behaviour and rules
Question code	Stable unique identifier	The conceptual question. Use a meaningful slug; do not reuse it for a different concept.
Version	1 for the first wording	Create a new version when changing a published question.
Question type	Single choice or True or false	The current learner UI expects one correct option.
Prompt	The full question	Avoid ambiguity and information not taught in the lesson.
Explanation	Correct-answer rationale	Learner feedback and remediation context.
Points	Positive whole number	Contributes to assessment score.
Remediation resource	Optional linked reading	Recommended when an incorrect answer should direct the learner to supporting material.
Is published	Select only after options are complete	Published question versions cannot be casually edited or deleted.
Answer label	Text displayed as an option	Use one row per answer.
Is correct	Select exactly one option	Course review blocks publication if a lesson check does not have exactly one correct option.
Answer order	1, 2, 3 and so on	Controls display order unless assessment randomisation changes question order.

Theory assessments

Assessment fields

Field	What to enter	Behaviour and rules
Programme version	The draft course version	All assessment items must support this course.
Module	Select for a module assessment; leave blank for final theory	A Final Theory Examination cannot belong to one module.
Title	Learner-facing assessment name	Use consistent names such as Module 2 Assessment or Final Theory Examination.
Assessment type	Module assessment or Final theory examination	Controls scope and unlocking behaviour.
Pass percentage	Required percentage such as 70	Used to calculate pass or fail.
Maximum attempts	Whole number or blank	Blank means unlimited attempts.
Randomize questions	Select when desired	Changes question order but does not alter the frozen question set for an attempt.
Is published	Select after items and answers are checked	The course publish action publishes connected assessments.
Assessment items	Published or ready Question Versions	Order controls the normal sequence. A question version can appear only once per assessment.

Final theory flow. The final theory examination remains locked until all published module lessons are complete. Time is recorded for information but does not affect pass or fail. Attempts are numbered and retained rather than overwritten.

Practical simulator flows

A practical contains teaching context plus a programmed sequence of simulator actions. The Simulator Workspace is the learner's sandbox: their actions create attempt records and do not change the course definition or other learners' work.

Guided practical builder

Field	What to enter	Behaviour and rules
Practical title	Name of the workflow being practised	Creates a Scenario and Scenario Version 1.
Area	Import, Export, Transit, or Warehouse	Controls classification and learner navigation.
Briefing	Fictitious assignment and starting situation	Tell the learner what has happened, their role, and the expected result.
Learning objective	Observable skill the learner should demonstrate	Used to explain why the practical exists.
Assistance mode	Beginner, Intermediate, or Advanced	Beginner includes hints; Intermediate has limited guidance; Advanced has no hints. Competency is configured in Advanced Admin.
Maximum attempts	Whole number or blank	Blank means unlimited attempts.
Steps	One simulated action per line, in execution order	Requires 2 to 30 steps. The builder creates states, transitions, completion effects, feedback, hints, and a completion rubric.

Advanced scenario version

Field	What to enter	Behaviour and rules
Scenario	Parent scenario identity	The title and area live on the Scenario record.
Module	Course module containing this practical	Links the practical into the course.
Version	1 initially	Create a new version instead of rewriting published history.
Status	Draft, Published, or Retired	Learners see published versions only.
Purpose	Practice, Practical module assessment, or Practical competency assessment	Practice does not qualify for a certificate. Competency can.
Assistance mode	Beginner, Intermediate, Advanced, or Competency	Purpose Competency requires Competency assistance mode.
Reference status	Conceptual, Verified, Adapted, or To be confirmed	Records the authority level of the simulated flow.
Initial data	Advanced JSON key/value state only	Leave as {} unless actions require flags. Keys are checked by conditions and changed by effects.

States and actions

Field	What to enter	Behaviour and rules
State key	Unique lowercase identifier	Unique inside the scenario version.
State label	Learner-facing stage name	Shown as the current practical stage.
Guidance	Instruction for this stage	Shown according to the assistance mode.
Is initial	Select on exactly one state	The learner attempt starts here.
Is terminal	Select on the completion state	Reaching it completes the practical.
Action code and label	Stable code plus learner-facing action	Code supports evidence; label is what the learner clicks.
From state and To state	Current and next state	Both must belong to the same scenario version. Blank states support parallel actions.
Conditions	Required JSON key/value pairs	The action appears only when the attempt state data satisfies them.
Effects	JSON key/value updates	Applied after the action, enabling later actions or recording evidence.
Success feedback	Confirmation shown after action	Tell the learner what changed.
Beginner hint	Next-step guidance	Available only where assistance permits it; assistance use can be evaluated.

Fictitious shipping documents

Create structured learner documents in Django Admin under Scenarios. Connect every document to the same Scenario Version used by the practical. Draft documents are author-only; published documents are available to learners who have an attempt for that scenario.

Bill of Lading

Use Guided form for entry, Live document view for immediate authoring preview, and View generated PDF after saving. Changing among Carrier Grid, Ocean Transport, and Multimodal changes presentation while retaining the entered data.

Bill of Lading document fields

Field	What to enter	Behaviour and rules
Scenario version	Practical receiving the BL	Determines learner access.
Status	Draft or Published	Publish only after checking the generated PDF.
Title and reference	Fictitious document name and unique BL number	Reference must be unique.
Template	Carrier Grid, Ocean Transport, or Multimodal	Changes layout, not shipment data.
Original status	For example Non-negotiable training copy	Printed document classification.
Number of originals	Usually 0 for simulator copies	Informational only.
Carrier and carrier agent	Fictitious shipping line and agent	Do not use a real transaction identity.
Shipper, consignee, notify party	Full fictitious party details	Keep identities consistent across invoice and packing list.
Booking and shipper references	Fictitious cross-references	Use consistent scenario identifiers.
Vessel and voyage	Fictitious vessel and voyage reference	Printed in the route section.
Receipt, loading, discharge, delivery	Route places in sequence	Place of receipt/delivery may differ from ports.
Freight terms	Prepaid, collect, or training wording	BL carriage term, not invoice line pricing.
Shipper declared value	Optional BL declaration	Not the commercial invoice total.
Issue and shipped dates	Fictitious but internally consistent dates	Shipment date should not follow issue events illogically.

BL cargo item fields and behaviour

Field	What to enter	Behaviour and rules
Cargo type	Vehicle, General merchandise, Household or personal effects, Machinery, or Other cargo	Controls which item fields appear.
Vehicle	Year, make, model, VIN/chassis, HS code, weight and measurement	Goods Description is hidden; vehicle identity fields are required.
General or personal effects	Goods Description plus packages and weights	Vehicle-specific fields are hidden.
Machinery	Description plus optional year, manufacturer, model, serial/chassis and HS code	Supports machinery with or without serial identity.
Other cargo	All available item fields	Use for mixed or unusual cargo.
Container and seal	Container identifier, seal, and type	Printed in the container/seal column.
Marks and numbers	Marks printed on packages or cargo	This identifies packages; it is not the narrative goods description.
Package quantity and type	Count and form, such as 20 cartons	Add another cargo item for each distinct vehicle or cargo line.
Gross weight and measurement	Numeric value plus unit	Printed per item and totalled in the PDF.

Commercial Invoice Proforma Invoice and Packing List

One editor handles all three. The selected type changes the heading, explanation, visible fields, live-preview columns, and generated PDF.

Type	Purpose	Pricing behaviour
Commercial Invoice	Final seller-issued commercial document	Shows unit price, line amount, FOB, Freight, Insurance, and C&F/CFR total.
Proforma Invoice	Quotation before purchase or shipment	Shows provisional prices and totals and prints a quotation-only warning.
Packing List	Physical shipment and package detail	Hides and rejects unit price, amount, FOB, Freight, and Insurance; shows packages, pieces, net weight, and gross weight.

Commercial document fields

Field	What to enter	Behaviour and rules
Scenario version and status	Target practical and Draft/Published	Published learner access requires a scenario attempt.
Document type	Commercial, Proforma, or Packing	Immediately changes wording, fields, live preview, and PDF structure.
Title, reference, date	Fictitious identity and date	Reference must be unique.
Exporter name/address/contact	Fictitious seller details	Printed in the header.
Consignee name/address	Fictitious buyer/importer	Keep consistent with the BL.
Currency	USD, EUR, GHS, etc.	Used for invoice and proforma prices.
Container reference	Container or proposed shipment reference	Optional but useful for matching documents.
Payment terms	Payment or quotation validity terms	For proforma, state validity and provisional nature.
FOB	Value of goods at export point	Invoice/proforma only. Defaults visually to line total when not entered.
Freight and Insurance	Separate commercial costs	Invoice/proforma only.
Description and quantity	One row per product or cargo item	Used by all document types.
Package count and pieces per package	For example 4 cartons and 5 pcs/carton	Emphasised on packing lists.
Net and gross weight	Weight excluding/including packaging	Shown separately on packing lists and totalled.
Dimensions and HS code	Optional supporting item details	Store useful detail even where the standard PDF is concise.
Unit price and amount	Price per unit and line total	Commercial/proforma only. Packing-list validation prevents prices.

Practical rubrics and competency assessment

The guided practical builder creates a demonstration completion rubric. A final competency practical needs deliberate Advanced Admin configuration and subject-matter review.

Rubric fields

Field	What to enter	Behaviour and rules
Rubric code and title	Stable identifier and readable name	Use a new version when scoring policy changes.
Scenario version	Exactly one practical version	Each practical rubric version belongs to one scenario version.
Status	Draft, Published, or Retired	Publishing the course publishes connected rubric versions.
Pass percentage	Required practical percentage	Mandatory criteria can still cause failure.
Is demonstration	Selected for unapproved practice rules	Clear this only after formal approval of competency policy.
Criterion title and description	The behaviour being evaluated	Make evidence observable.
Dimension	Accuracy, review, procedure, decision, correction, assistance, or completion	Groups the skill measured.
Maximum points	Positive whole number	Contributes to total practical score.
Mandatory	Select when failure must fail the practical	Overrides a high overall percentage when unmet.
Evaluation rule	Completion, required_actions, state_flags, or assistance_limit JSON	Advanced field. The rule reads saved attempt evidence.
Remediation resource or scenario	Follow-up support	Presented when the criterion is not met.

Publishing and enrolment

Course review gate

Review course blocks publishing when any module has no lesson, any lesson has no teaching content, any lesson has no knowledge check, a lesson check lacks exactly one correct answer, or a practical lacks at least two states and an action.

Before selecting Publish course, manually confirm the final theory assessment, learner documents, practical purpose, competency rubric, completion policy, and realistic cross-document consistency. These advanced checks are not all enforced by the guided review gate.

Create and enrol a student

Field	What to enter	Behaviour and rules
Username, password, name, email	The student's account details	Use the Users area in Django Admin.
Groups	Student and any authorised additional role	Creating an enrolment through the user page automatically adds Student.
Programme version	The published course version	Do not enrol into a draft course for normal learning.
Enrolment status	Active, Completed, or Withdrawn	Active grants the learner journey. Completion is normally system-driven.
Enrolled by	Recorded administrator	Filled automatically when the enrolment is created in the user page.

Learner flow and final completion

Stage	Unlock or completion rule	Administrative evidence
Disclaimer	Learner accepts the current disclaimer	Disclaimer acceptance and audit event.
Theory lessons	Learner reads published lessons and completes checks	Lesson progress and lesson check responses.
Module assessment	Available for its module according to course flow	Frozen assessment attempt, responses, score, and outcome.
Final theory	Unlocks after all published module lessons are complete	Submitted passing final-theory attempt.
Orientation	Unlocks after final theory pass	Orientation completion record.
Practical	Unlocks after orientation	Scenario attempt, state data, action history, and assistance events.
Practical evaluation	Generated from the frozen rubric and action evidence	System score/outcome plus any audited instructor revision.
Certificate	Requires orientation, passed final theory, and passed competency-purpose practical	Completion record and certificate. Practice-only scenarios do not qualify.

Advanced Django Admin reference

Area	Create or configure	Treat as read only or protected evidence
Accounts	Users and role groups	Authentication history is audited.
Onboarding	Programmes, draft programme versions, enrolments, disclaimer versions	Disclaimer acceptances.
Learning	Modules, lessons, content blocks, resources	Published historical content should be versioned, not rewritten.
Assessments	Questions, question versions, options, assessments and items	Theory attempts, responses, lesson-check responses.
Scenarios	Scenarios, versions, states, actions, structured documents	Attempts and action history.
Evaluations	Rubrics, versions, criteria, remediation	Practical evaluation, criterion results, revisions and feedback.
Reports	Completion policy	Completion records and certificates.
Audit	No normal content creation	Append-only security and administrative events.

Versioning rule. Draft records may be edited. Once content, a question, assessment, practical, rubric, or programme version has been published and used, create a new version for substantive changes so earlier learner evidence remains meaningful.

Final quality checklist and troubleshooting

- The course remains Draft while being assembled.
- Every module contains at least one lesson.
- Every lesson contains teaching content and a knowledge check.
- Every knowledge check has exactly one correct answer and an explanation.
- Module assessments contain the intended question versions.

- The final theory assessment belongs to the programme version and has no module selected.
- Each guided practical has a clear briefing, objective, ordered steps, initial state, terminal state, and valid transitions.
- Competency scenarios use Competency purpose and Competency assistance mode.
- Each practical has an appropriate rubric and approved mandatory rules.
- BL, invoice/proforma, and packing-list party, reference, quantity, package, weight, and route details agree.
- Every structured document PDF has been opened and visually checked before publication.
- Course Review shows no blocking issues.
- A test student can complete the full journey in the expected order.

Common problems

Problem	Likely reason	Action
Theory page returns 403	User lacks an active enrolment, Student access, or the content is not published	Check user groups, active enrolment, programme status, and lesson/assessment publication.
Course cannot publish	Review gate found missing module, lesson, content, question, correct option, or practical flow	Open Review course and resolve each listed item.
Final theory is locked	Published lessons are incomplete	Complete the theory roadmap using a student account; do not manually alter evidence.
Practical is locked	Final theory or orientation is incomplete	Complete the preceding learner stages.
Document PDF returns 404	Legacy document reference was removed or URL targets the wrong record type	Use the structured document's own View generated PDF link after saving.
Packing list shows price fields	Old static assets are cached or wrong document type selected	Refresh the admin page and select Packing List; save and regenerate.
Live preview is unavailable	The record is on the legacy Scenario Document editor	Use Bills of Lading or Commercial Documents under Scenarios.
No certificate after practical	The practical purpose is Practice or competency evidence is incomplete	Require a passed competency-purpose practical plus passed final theory and orientation.

Safe administration principles

- Use fictitious identities, references, values, vehicles, and transactions in learner content.
- Never represent simulator output as an official customs, banking, shipping, or commercial document.
- Keep Django Admin restricted to trusted staff and do not expose a development server to the public internet.
- Review audit records before correcting access, enrolment, evaluation, or completion disputes.
- Preserve published history by creating new versions instead of modifying evidence already used by learners.